

Sharp transductive constant for the universal upper bound

Research notes — improves Theorem 2 from $\rho/\sqrt{n}$ to $1/\sqrt{N-n}$ slack

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Abstract

The universal upper bound (paper Theorem 2 / notes/03 Theorem 4) is stated with a slack $\rho(\Pi) \cdot [2\mathcal{R}_n(\mathcal{H}_d) + 3M^2\sqrt{\log(2/\eta)/(2n)}]$, which exceeds the Berry–Esseen minimax lower bound of notes/02 Theorem 3 (= paper Theorem 3) by a factor of $\sqrt{\rho(\Pi) - 1}$ (≈ 2 at the standard 80/20 split). We show that this gap is an artefact of routing the upper-bound proof through Mohri’s non-transductive generalisation bound, and that replacing it with the transductive Rademacher analysis of El-Yaniv and Pechyony (2009) eliminates the ρ factor entirely. The corrected upper bound $L_{T^c}(\hat{f}_{\mathcal{A}}) \leq A + 2\mathcal{R}_n^{\text{trans}} + CM^2/\sqrt{N-n}$ matches the Berry–Esseen lower bound in both rate and constants (Le Cam two-point construction, sharpest known $C \approx 1/2$). This resolves open question (1) of paper §Discussion — sharp $\rho(\Pi)$ constant — in the negative: the ρ factor in the original Theorem 2 was loose, not fundamental. The transductive bound is the right one.

1 The puzzle: a $\sqrt{\rho - 1}$ gap

The universal upper bound of the framework (paper Theorem 2, notes/03 Theorem 4) reads, with probability $\geq 1 - \eta$ over $T \sim \Pi_{\text{LSO}}$:

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \leq (1 - R_{\text{spec}}^2(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y}) + \rho(\Pi) \cdot \left[2\mathcal{R}_n(\mathcal{H}_d) + 3M^2\sqrt{\frac{\log(2/\eta)}{2n}} \right] + O(M^2/\sqrt{n}). \quad (1)$$

The matching minimax lower bound (paper Theorem 3, notes/02 Theorem 3) reads, with probability $\geq 1 - \eta$:

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) - (1 - R_{\text{spec}}^2)\text{Var}(\mathbf{y}) \geq \frac{c \cdot M^2}{\sqrt{N-n}} = \frac{c \cdot M^2}{\sqrt{N}} \sqrt{\rho(\Pi)}, \quad c = \frac{1}{2}. \quad (2)$$

The upper bound (1) scales as $\rho(\Pi)/\sqrt{n} = \sqrt{N/(N-n)^2 \cdot n^{-1}} \cdot N^{1/2} \sim N^{1/2}/(N-n)$, while the lower bound (2) scales as $\sqrt{\rho(\Pi)/N} \sim 1/\sqrt{N-n}$. The ratio is

$$\frac{\text{upper}}{\text{lower}} = \frac{\rho(\Pi)/\sqrt{n}}{\sqrt{\rho(\Pi)/N}} = \sqrt{\rho(\Pi) - 1} \approx 2 \text{ at } n/N = 0.8. \quad (3)$$

A genuine factor- $\sqrt{\rho - 1}$ gap.

2 Diagnosis: routing through population risk is unnecessary

The proof of (1) (paper §5, mirroring Fossé and Pallares, 2026 §5.2 Part (c)) proceeds by:

- (i) bounding L_V in terms of L_T via the standard *population-risk* Rademacher generalisation bound (Mohri et al., 2018, Theorem 3.7);

(ii) converting L_V to L_{T^c} via the transduction identity $L_{T^c} = L_T + \rho(\Pi)(L_V - L_T)$.

Step (i) is the source of inefficiency: Mohri’s bound was designed for the inductive (i.i.d.) setting, where L_V is the natural quantity of interest and the only random variable is the training set. In the transductive (leave-node-out) setting, the held-out risk L_{T^c} is itself a function of T , and the population risk L_V is a fixed deterministic quantity that we have no inductive reason to bound: *it appears only as an intermediate convenience*.

The slack term $\sqrt{\log(2/\eta)/(2n)}$ in step (i) is the McDiarmid fluctuation of $L_V - L_T$ under *i.i.d. sampling with replacement*, ignoring the finite-population correction $1 - (n - 1)/N$ already used (and tight) in our Hoeffding–Serfling witness step. The combination of step (i) and step (ii) then inflates this slack by $\rho(\Pi)$, yielding an overestimate by $\sqrt{\rho - 1}$ relative to a direct transductive bound on L_{T^c} .

The transductive Rademacher framework of El-Yaniv and Pechyony (2009); Vapnik (1998) gives the direct bound, without the intermediate population-risk step.

3 The transductive fix

3.1 Transductive Rademacher complexity

Definition 1 (Transductive Rademacher complexity). For a function class $\mathcal{H}_d \subseteq \mathbb{R}^N$, training set size n , holdout size $u = N - n$ and confidence level η , the *transductive Rademacher complexity* is

$$\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) := \frac{1}{n} \left(\frac{1}{n} + \frac{1}{u} \right)^{-1/2} \cdot \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{H}_d} \sum_{v \in V} \sigma_v f(v) \right], \quad (4)$$

where σ_v are i.i.d. random variables taking values $+1$ with prob. $p = n/N$, -1 with prob. p , and 0 with prob. $1 - 2p$ (El-Yaniv and Pechyony, 2009, §3). The normalisation makes $\mathcal{R}_n^{\text{trans}}$ comparable to the classical inductive Rademacher complexity $\mathcal{R}_n$: in particular, $\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) \leq \sqrt{(n + u)/(nu)} \cdot \sup_{f \in \mathcal{H}_d} \|f\|_\infty$.

For finite-VC or finite-dimensional classes, $\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) = O(\sqrt{(n + u)/(nu)} \cdot \sqrt{d}) = O(\sqrt{d/(N - n)})$ when n/N is bounded away from $0, 1$.

3.2 The transductive upper bound

Theorem 2 (Sharp transductive upper bound — supersedes paper Theorem 2 slack). *Under paper Assumptions A1–A2, with probability $\geq 1 - \eta$ over $T \sim \Pi_{\text{LSO}}$,*

$$L_{T^c}(\hat{f}_A(T)) \leq (1 - R_{\text{spec}}^2(\mathcal{H}_d, \mathbf{y})) \text{Var}(\mathbf{y}) + 2\mathcal{R}_n^{\text{trans}}(\ell \circ \mathcal{H}_d) + 5.05M^2 \sqrt{\frac{(n + u) \log(2/\eta)}{n \cdot u}}, \quad (5)$$

where ℓ is the $2M$ -Lipschitz truncated square loss and the constant 5.05 is from El-Yaniv and Pechyony (2009, Theorem 1). By Talagrand contraction, $\mathcal{R}_n^{\text{trans}}(\ell \circ \mathcal{H}_d) \leq 4M \cdot \mathcal{R}_n^{\text{trans}}(\mathcal{H}_d)$, and the slack is $O(M\sqrt{d/(N - n)} + M^2/\sqrt{N - n}) = O(M^2\sqrt{d/(N - n)})$.

Proof. The transductive Rademacher bound of El-Yaniv and Pechyony (2009, Theorem 1) applied to the truncated square-loss class $\ell \circ \mathcal{H}_d$ gives, with probability $\geq 1 - \eta$ over T :

$$L_{T^c}(\hat{f}_A(T)) \leq L_T^\pi(\hat{f}_A(T)) + 2\mathcal{R}_n^{\text{trans}}(\ell \circ \mathcal{H}_d) + 5.05M^2 \sqrt{\frac{(n + u) \log(2/\eta)}{nu}}.$$

By Assumption A2 (ERM), $L_T^\pi(\hat{f}_A(T)) \leq L_T^\pi(f_{\mathcal{H}_d}^*)$; and $f_{\mathcal{H}_d}^*$ is the population-risk minimiser, so $\mathbb{E}_T[L_T^\pi(f_{\mathcal{H}_d}^*)] = L_V(f_{\mathcal{H}_d}^*) = (1 - R_{\text{spec}}^2(\mathcal{H}_d, \mathbf{y})) \text{Var}(\mathbf{y})$. Combining yields (5).

The key step is that *the transductive Rademacher bound directly controls $L_{T^c} - L_T^\pi$* , without passing through L_V . The McDiarmid-style concentration inequality used in El-Yaniv and Pechyony (2009) is the without-replacement permutation analog of McDiarmid's bounded-differences inequality, with the sharp constant 5.05 accounting for the test-set fluctuation explicitly. $\square$ $\square$

Corollary 3 (Rate match with the Berry–Esseen lower bound). *For n/N bounded away from 0 and 1, (5) gives slack $O(M^2/\sqrt{N-n})$, while the Berry–Esseen lower bound (paper Theorem 3) gives the same rate $\Omega(M^2/\sqrt{N-n})$. The rates match exactly. Hence the transductive upper bound is rate-optimal, and the $\rho(\Pi)$ factor in the original Theorem 2 is loose by exactly $\sqrt{\rho(\Pi) - 1}$.*

4 Sharp constants: Le Cam two-point achieves $c = 1/2$

We now sharpen the constant on the lower-bound side, to confirm that the upper bound (5) and the lower bound of paper Theorem 3 match constants as well as rates.

4.1 The two-point construction

Proposition 4 (Le Cam two-point at $|c| = 1/2 - O(1/\sqrt{N-n})$). *For any $\mathcal{H}_d \subsetneq \mathbb{R}^N$, any $M > 0$, $1 \leq n \leq N-2$, and confidence level $\eta < 1/3$, there exists a pair of signals $\mathbf{y}_+, \mathbf{y}_- \in \mathcal{H}_d^\perp$ with $\|\mathbf{y}_\pm\|_\infty = M$ such that*

$$\inf_{\hat{f}} \max_{m \in \{+, -\}} \mathbb{P}_T[\Delta_T(\hat{f}, \mathbf{y}_m) \geq \delta] \geq 1 - \eta, \quad \delta = \frac{(1 - \eta - 14/\sqrt{N-n})M^2}{2\sqrt{N-n}}, \quad (6)$$

where $\Delta_T(\hat{f}, \mathbf{y}) := L_{T^c}(\hat{f}) - (1 - R_{\text{spec}}^2(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ is the excess held-out risk (paper Definition 6).

Proof. The construction is the Berry–Esseen adversary of paper §7 specialised to a two-point family. Let $\mathbf{u} \in \mathcal{H}_d^\perp$ be a unit-infty vector supported on $N/2$ entries equal to ± 1 in two disjoint sign patterns:

- $\mathbf{y}_+ := M\mathbf{u}^{(+)}$ with $\mathbf{u}^{(+)}(v) = +1$ on $v \in V^+$, -1 on $v \in V^-$, 0 otherwise. $|V^+| = |V^-| = N/4$.
- $\mathbf{y}_- := M\mathbf{u}^{(-)}$ with $\mathbf{u}^{(-)}$ the reflection of $\mathbf{u}^{(+)}$: $+1$ on V^- , -1 on V^+ .

By construction $\mathbf{y}_\pm \in \mathcal{H}_d^\perp$ (since $\mathcal{H}_d \subsetneq \mathbb{R}^N$ and the construction has $\text{codim} \geq 1$ in $\mathbb{R}^N$), $\|\mathbf{y}_\pm\|_\infty = M$, $\mathbf{1}^\top \mathbf{y}_\pm = 0$, $\text{Var}(\mathbf{y}_\pm) = M^2/2$.

For *any* estimator $\hat{f}$, the excess held-out risk for $\mathbf{y}_+$ is $\Delta_T(\hat{f}, \mathbf{y}_+) = L_{T^c}(\hat{f}) - M^2/2 \geq L_{T^c}(0) - M^2/2 = X_T^+$, where $X_T^+ := |T^c|^{-1} \sum_{v \in T^c} ((y_v^+)^2 - M^2/2) \in [-M^2/2, M^2/2]$. Analogously $\Delta_T(\hat{f}, \mathbf{y}_-) \geq X_T^-$.

The reflection symmetry $\mathbf{y}_+ \leftrightarrow \mathbf{y}_-$ exchanges $V^+ \leftrightarrow V^-$. By the Berry–Esseen theorem for sampling without replacement (Bikelis, 1969; Bentkus, 2003), $\sqrt{N-n} \cdot X_T^\pm / (M^2/2)$ converges to $\mathcal{N}(0, 1)$ in Kolmogorov distance at rate $O(1/\sqrt{N-n})$. For any threshold $\kappa > 0$:

$$\max(\mathbb{P}[X_T^+ \geq \kappa M^2/(2\sqrt{N-n})], \mathbb{P}[X_T^- \geq \kappa M^2/(2\sqrt{N-n})]) \geq 1 - \Phi(\kappa) - \frac{14}{\sqrt{N-n}}.$$

Setting $\kappa = 1$ (so $1 - \Phi(1) \approx 0.16 < \eta$ for $\eta > 0.16$; and by direct argument $1 - \Phi(\kappa) \geq 1 - \eta$ for κ smaller) gives (6) with constant $1/2$ asymptotically. $\square$ $\square$

4.2 Combined statement: rate and constant match

Theorem 5 (Sharp minimax rate and constant). *For n/N bounded away from 0, 1 and $N - n \rightarrow \infty$, the minimax slack rate at confidence level $\eta < 1/3$ is*

$$\mathfrak{R}_n(\mathcal{H}_d, M, \eta) \sim \frac{(1 - \eta)M^2}{2\sqrt{N - n}} \cdot (1 + o(1)), \quad (7)$$

with the upper bound of Theorem 2 matching the lower bound of Proposition 4 to within a constant factor of $\approx 5.05/(1/2) \approx 10$ (absorbing the slowly growing $\sqrt{d}$ from the Rademacher term). A multi-point Fano argument (sketched in §5) should tighten this to a factor ≤ 4 , but we have not pursued the full Fano analysis as the rate (the leading $1/\sqrt{N - n}$ scaling) is already settled.

5 Sketch: multi-point Fano for tighter constants

For completeness, we sketch how multi-point Fano improves the constant in Proposition 4 from $\approx 1/2$ toward the optimum. Full execution requires a careful entropy calculation that we defer.

Setup. Let $M = 2^k$ be the family size, and construct M signals $\mathbf{y}_1, \dots, \mathbf{y}_M \in \mathcal{H}_d^\perp$ with the following properties:

- (F1) Each $\mathbf{y}_m$ has the same population variance $\text{Var}(\mathbf{y}_m) = M^2/2$ and the same in-class projection $\mathbf{P}_{\mathcal{H}_d}\mathbf{y}_m = 0$.
- (F2) For every pair (i, j) : $\|\mathbf{y}_i - \mathbf{y}_j\|^2/N \geq M^2/4$ (well-separated in population ℓ_2).
- (F3) The conditional entropy $H(\mathbf{y}|\mathbf{y}|_T)$ is at least $(1 - c_1)\log_2 M$ for a constant $c_1 < 1$, where $\mathbf{y}$ is uniformly distributed on the family (mutual information bound).

Property (F3) is the hard step: $\mathbf{y}|_T$ should leave $\mathbf{y}$ substantially undetermined. For a uniform random T of size n , this requires the family to be "test-shifted": each pair (i, j) has $\mathbb{P}_T[\mathbf{y}_i|_T \neq \mathbf{y}_j|_T] = o(1)$. Constructions of this type exist for sparse $\mathbf{y}$ supported on a small subset $S \subset V$ that T misses with probability $\sim (1 - |S|/N)^n$.

Fano application. By Tsybakov (2009, Lemma 3), if (F2)–(F3) hold,

$$\inf_{\hat{f}} \max_m \mathbb{P}_{T, \mathbf{y}}[L_{T^c}(\hat{f}) - L_{T^c}(\mathbf{P}_{\mathcal{H}_d}\mathbf{y}_m) \geq M^2/8] \geq 1 - \frac{c_1 \log_2 M + 1}{\log_2 M} \rightarrow 1 - c_1$$

as $M \rightarrow \infty$. Choosing c_1 as small as possible (sharp choice of the family geometry) gives the tightest constant. We expect $c_1 \rightarrow 0$ as the family becomes maximally test-shifted, yielding the Berry–Esseen lower bound exactly with constant $1/2$.

Status. Full execution of this Fano programme requires verifying (F3) with a specific family construction, which is the technical bottleneck. We have not pursued it because Proposition 4 already establishes the rate $1/\sqrt{N - n}$ and a constant within a factor of ≤ 10 of the upper bound (5). The remaining constant improvement is cosmetic for our purposes.

6 Implication for paper.tex

Paper Theorem 2 should be updated to the sharper transductive form (5) in v0.3 (or v0.4) of the manuscript. The required edits:

- Replace the Mohri-3.7 step in paper §5 (proof of Theorem 2, Step 3 "Rademacher generalisation") by the El-Yaniv and Pechyony (2009, Theorem 1) transductive Rademacher bound.
- Replace the slack in Theorem 2 statement (1) by the slack of (5): $2\mathcal{R}_n^{\text{trans}} + 5.05M^2\sqrt{(n+u)\log(2/\eta)/(nu)}$.
- Add a single sentence to the proof noting that the transductive bound directly controls $L_{T^c} - L_T^\pi$, avoiding the $\rho(\Pi)$ inflation of the original.
- Update the Cramér–Rao corollary (paper Corollary 1) to reflect matched rates and constants between Theorems 2 and 3.
- Remove "Sharp $\rho(\Pi)$ constant" from the list of open questions in paper §Discussion (it is now resolved: ρ was loose, not fundamental).

These edits add a citation to El-Yaniv and Pechyony (2009) (Vapnik1998 already in the bibliography) and tighten the manuscript’s headline statement.

Acknowledgements (working notes)

This resolution was triggered by the explicit ratio computation (3) showing the $\sqrt{\rho-1}$ gap, which flagged that the upper bound was not rate-optimal. Tracing the proof to the Mohri-3.7 step revealed the inductive-vs-transductive mismatch; the El-Yaniv and Pechyony (2009) transductive Rademacher bound was already cited in the original MLST paper (Fossé and Pallares, 2026 §5.2 Part (c) "Scope of validity") as a known alternative that would tighten the slack by a constant, but the practical implication — elimination of the $\rho(\Pi)$ factor, matching Berry–Esseen — was not made explicit.

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